Apply filters to SQL queries

Project description

[To investigate security issues and help keep the system secure. I accomplish this by examining data in their employees and log_in_attempts tables using SQL filters. I will be using the logical operators AND, OR, and NOT to retrieve records.]

Retrieve after hours failed login attempts

[`SELECT * FROM log_in_attempts WHERE login_time > '1800';`] This query instructs SQL to return all columns from the log in attempts table with the greater than operator as the condition for the filter to show all login times greater than 1800. With the data returned, we can review the after hours log in attempts and identify all failed log in attempts occurring after 1800 with a value of 0 or false.

Retrieve login attempts on specific dates

[`SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';`] This query instructs SQL to return all columns from the log in attempts table with OR as the condition for the filter to specify that either condition can be met. In this case the conditions are log in attempts occurring on either 2022-05-09 or 2022-05-08. This enables me to review and identify all log in attempts on that day and the day before if a suspicious event occurred.

Retrieve login attempts outside of Mexico

[`SELECT * FROM log_in_attempts WHERE NOT country LIKE 'MEX%';`] This query instructs SQL to return all columns from the log in attempts table with NOT, LIKE, and the percentage sign as the condition for the filter to substitute any number of other characters to negate both values of MEX and MEXICO. The LIKE operator is used here to search for this pattern in the country column. This enables me to identify and investigate all log in attempts occurring outside of Mexico for suspicious activity.

Retrieve employees in Marketing

[`SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';`] This query instructs SQL to return all columns from the employees table with equal to, AND, LIKE, and the percentage sign as the condition for the filter to specify to include both the department column which contains values equal to include Marketing and the office column to substitute any number of other characters in the column for the values of the East building. The LIKE operator is used here to search for this pattern in the office column. This enables me to identify all employees in the marketing department for all offices in the East building.

Retrieve employees in Finance or Sales

[`SELECT * FROM employees WHERE department = ‘Sales’ OR department = ‘Finance’;`] This query instructs SQL to return all columns from the employees table with equals to, and OR as the condition for the filter to specify that either condition can be met. In this case the conditions are the department column containing the values that include Sales and Finance. This enables me to identify all employees in the Sales or Finance department.

Retrieve all employees not in IT

[`SELECT * FROM employees WHERE NOT department = ‘Information Technology’;`] This query instructs SQL to return all columns from the employee table with NOT, and equals to as the condition to negate the department column containing values that include Information Technology. This enables me to identify all employees not in the information technology department.

Summary

[As a security professional at a large corporation, I investigated security issues and examined the organization’s data and discovered potential security issues. I retrieved failed login attempts for after hours, login attempts on specific dates, login attempts outside of Mexico, retrieved employees in marketing, finance, sales, and outside of the information technology department. Issues involving login attempts and employee machines were examined using logical operators such as AND, OR, and NOT to filter through the data along with other operators such as greater than, equal to, and LIKE with wildcards like the percentage sign to give conditions for the filters and search for patterns.]